Querying SharePoint Data in Power BI

Options & Performance

September 15th, 2026





About Me



John Kerski

Microsoft Most Valuable Professional

Experience

- 12+ years with Microsoft BI stack
- 10+ years with Power BI
- 10+ years with SharePoint

Credentials

- MCSE: Productivity
- MS: PBI Data Analyst
- PMP

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Agenda

- Lists
  - Options
  - Performance
- Document Libraries
  - Options
  - Performance

SharePoint Lists

SharePoint Online List Version 1




SharePoint
V1 Connector
Power BI

With larger lists…



V1 - Pros & Cons

Pros

- Easy To Use
- Good for small lists (< 5,000 items)

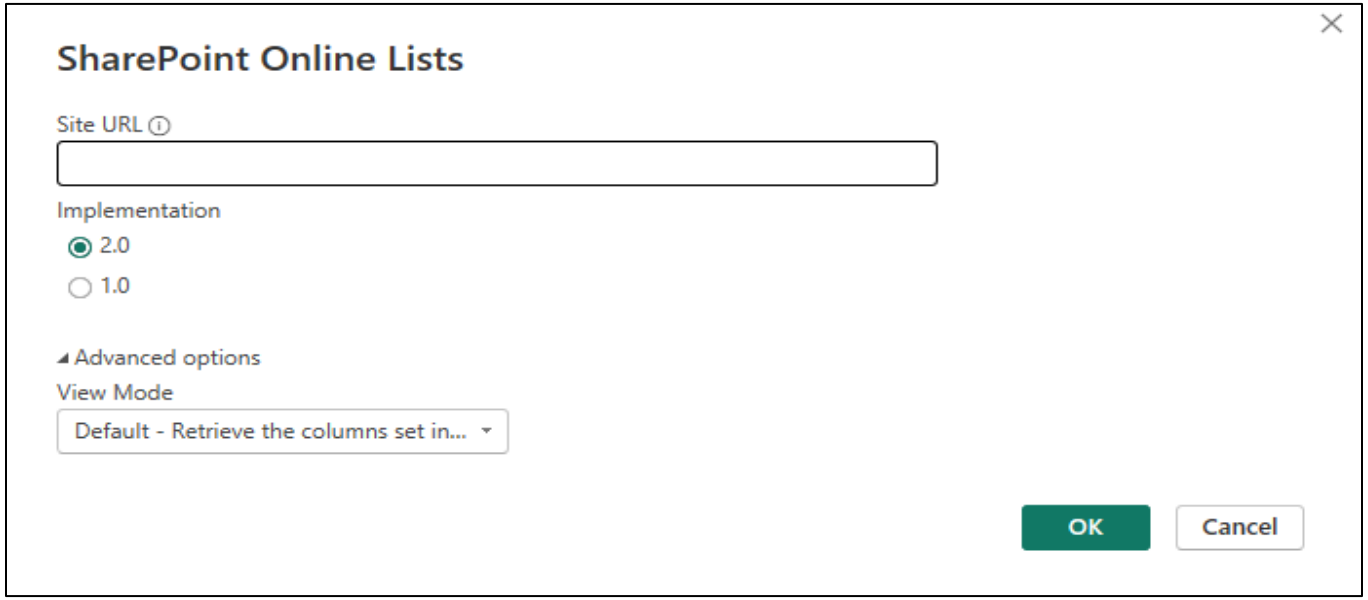
Cons

- Get More Data Than Needed
- Lots of “Cleanup” Power Query Steps
  - Remove Columns
  - Expand Columns
- Slow for larger lists
  - Especially with several lookups

Roche's Maxim on Data Transformation

Data should be transformed as far upstream as possible, and as far downstream as necessary

SharePoint Online List Version 2



V2 - Pros & Cons

Pros

- Easy To Use
- Good for small lists (< 5,000 items)

Cons

- Public View dependency could lead to schema drift
  - Or Errors
- Slow for larger lists
  - Especially with several lookups

OData


OData feed
Basic
Advanced
URL
http://services.odata.org/Northwind/Northwind.svc
OK
Cancel

OData - Example

firstname	gender	sentence	bool	ID	Modified	Modified By
Delbert	Male	Cum sociis natoque penatibus et magnis dis parturient montes, nascet...	TRUE	1	2024-02-27T20:54:03Z	Record
Nell	Female	Curabitur in libero ut massa volutpat convallis. Morbi odio odio, eleme...	TRUE	2	2024-02-27T20:54:03Z	Record
Khalil	Male	Morbi vestibulum, velit id pretium iaculis, diam erat fermentum justo, ...	TRUE	3	2024-02-27T20:54:03Z	Record
Gordan	Male	Morbi vel lectus in quam fringilla rhoncus. Mauris enim leo, rhoncus se...	TRUE	4	2024-02-27T20:54:03Z	Record
Hayyim	Male	Morbi a ipsum. Integer a nibh. In quis justo.	FALSE	5	2024-02-27T20:54:03Z	Record
Jordan	Male	Donec quis orci eget orci vehicula condimentum.	FALSE	6	2024-02-27T20:54:04Z	Record
Currey	Male	In est risus, auctor sed, tristique in, tempus sit amet, sem. Fusce conse...	FALSE	7	2024-02-27T20:54:04Z	Record
Deane	Male	Suspendisse potenti. Nullam porttitor lacus at turpis. Donec posuere ...	TRUE	8	2024-02-27T20:54:04Z	Record
Tracy	Female	Nulla ut erat id mauris vulputate elementum. Nullam varius. Nulla facil...	FALSE	9	2024-02-27T20:54:04Z	Record
Tobiah	Male	Nam congue, risus semper porta volutpat, quam pede lobortis ligula, s...	FALSE	10	2024-02-27T20:54:04Z	Record

/_api/lists/GetByTitle('SampleData')/items?$select=field_1,field_3,field_6,field_7,ID,Modified,Editor/EMail,Editor/Title&$expand=Editor&$top=5000

OData - Pros & Cons

Pros

- Good for large lists

Cons

- You must learn OData syntax: Working with lists and list items with REST | Microsoft Learn
- Defaults to SharePoint Paging Limit
- 100-page limit
  - 5,000 items = 50 network calls
- Credentials required for each call

Custom-made (Web.Contents)

fnGetFields

- *You*
  - Supply list name
- *Function*
  - Contacts the /fields endpoint in SharePoint to retrieve the pertinent columns about the SharePoint list.

Table of Fields

- *You*
  - Filter fields furthermore to refine query.

fnGetSharePointData

- You
  - Supply the Table of Fields
  - Provide any additional filters.
- Function
  - Converts Fields into Odata query.
  - Automatically expand results.

[An alternative way to import SharePoint list data into Power BI. (github.com)]

[Share a query - Power Query | Microsoft Learn]

Results on a Fabric Capacity (> 10,000 items)

Dataset	Median – Seconds to Refresh	Standard Deviation – Seconds to Refresh	Capacity Units	Number of Refreshes
V2	151.50	41.98	5,238.66	30
V1	103.20	8.36	5,053.90	30
OData	37.47	7.22	3,197.31	30
Custom-made	18.93	2.81	1,151.65	30

Custom-made - Pros & Cons

Pros

- Good for large lists
- You don’t have to learn OData syntax
- You can use template to generate OData for you

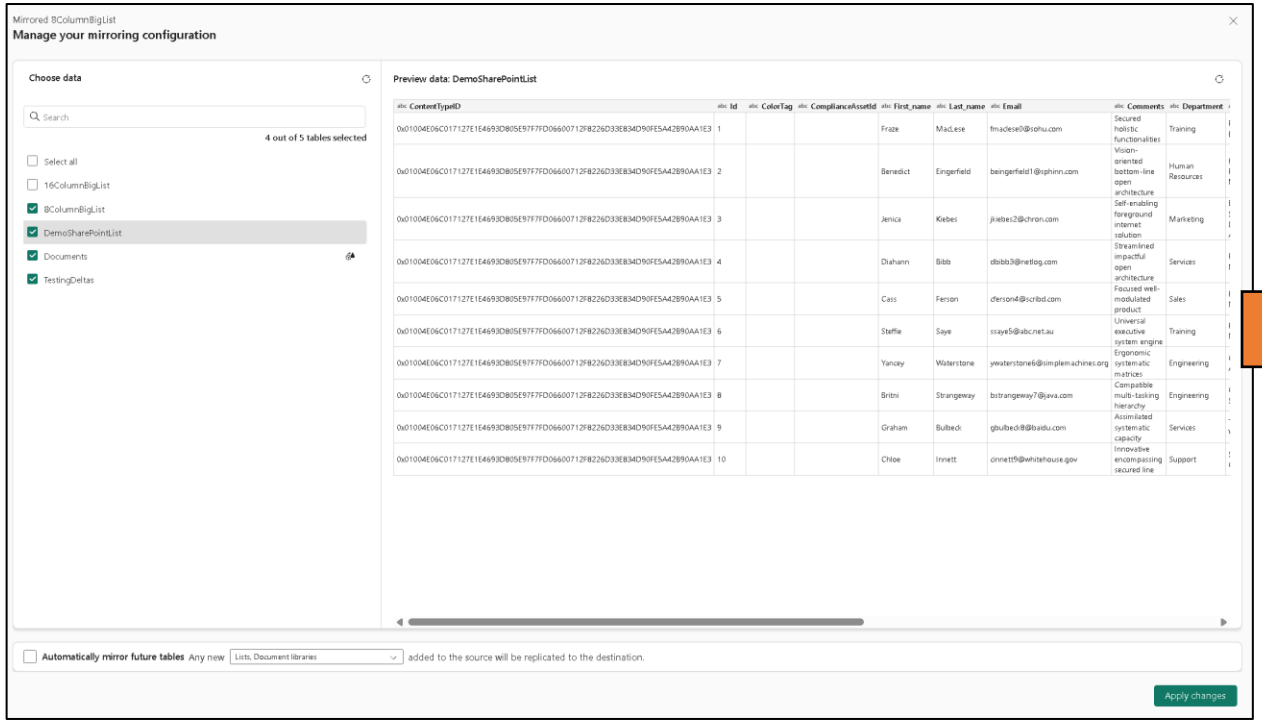
Cons

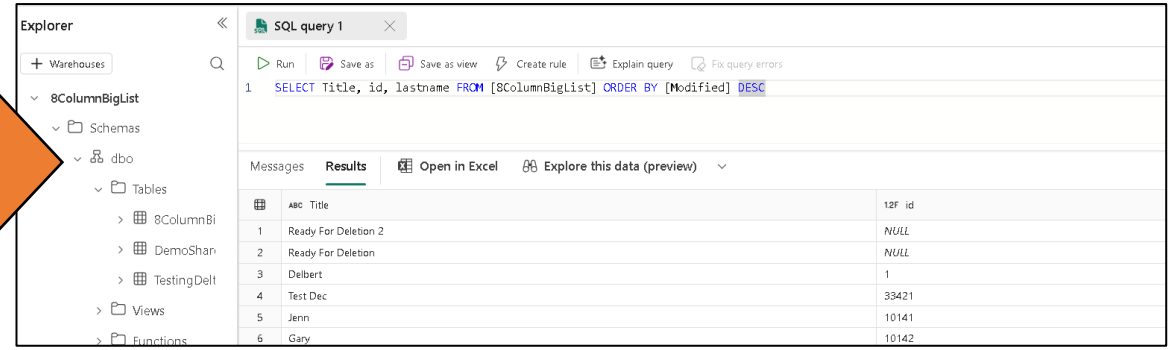
- Combining Data from different SharePoint site may conflict with privacy settings
  - Alternative: Do this in dataflows.

SharePoint Lists – Best Practices

- Avoid V2
- New to Power Query
  - Consider OData for Large Lists
- Experienced with Power Query
  - Switch to Custom-made approach

SharePoint List Mirroring





SharePoint List Mirroring - Pros & Cons

Pros

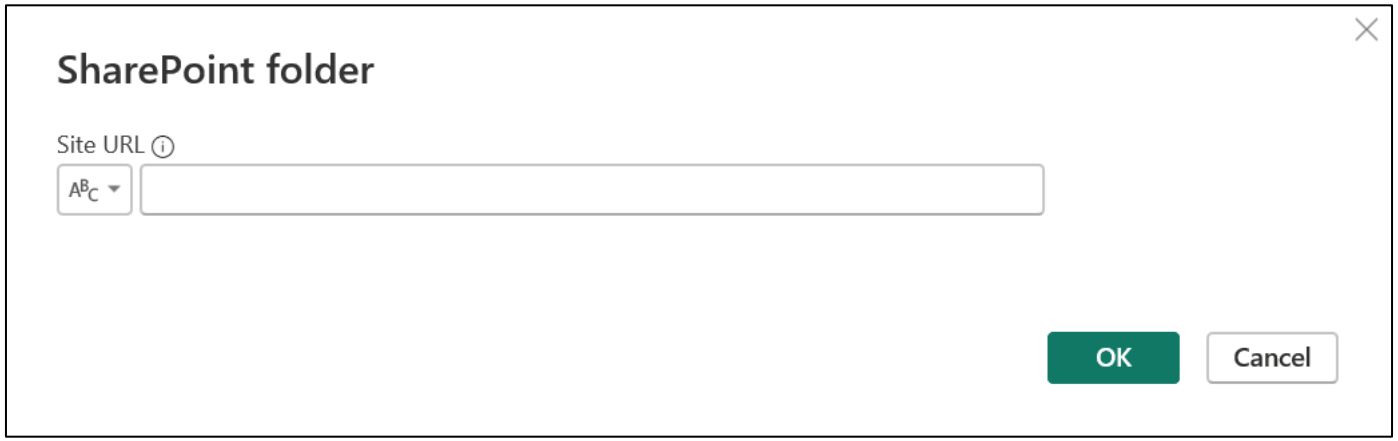
- No Manual Pipelines
- Updated every 2 minutes
- Allows you to process outside of the semantic model
- Can automatically pick up new lists

Cons

- In Preview (Sept. 2026)
- Need Fabric Licensing
  - Unsupported in GCC/DoD
- Slower if not processing outside of semantic model
- Tracking Deletes

SharePoint Document Libraries

SharePoint Folder (a.k.a. SharePoint.Files)



You want SharePoint files?



SharePoint.Files - Pros & Cons

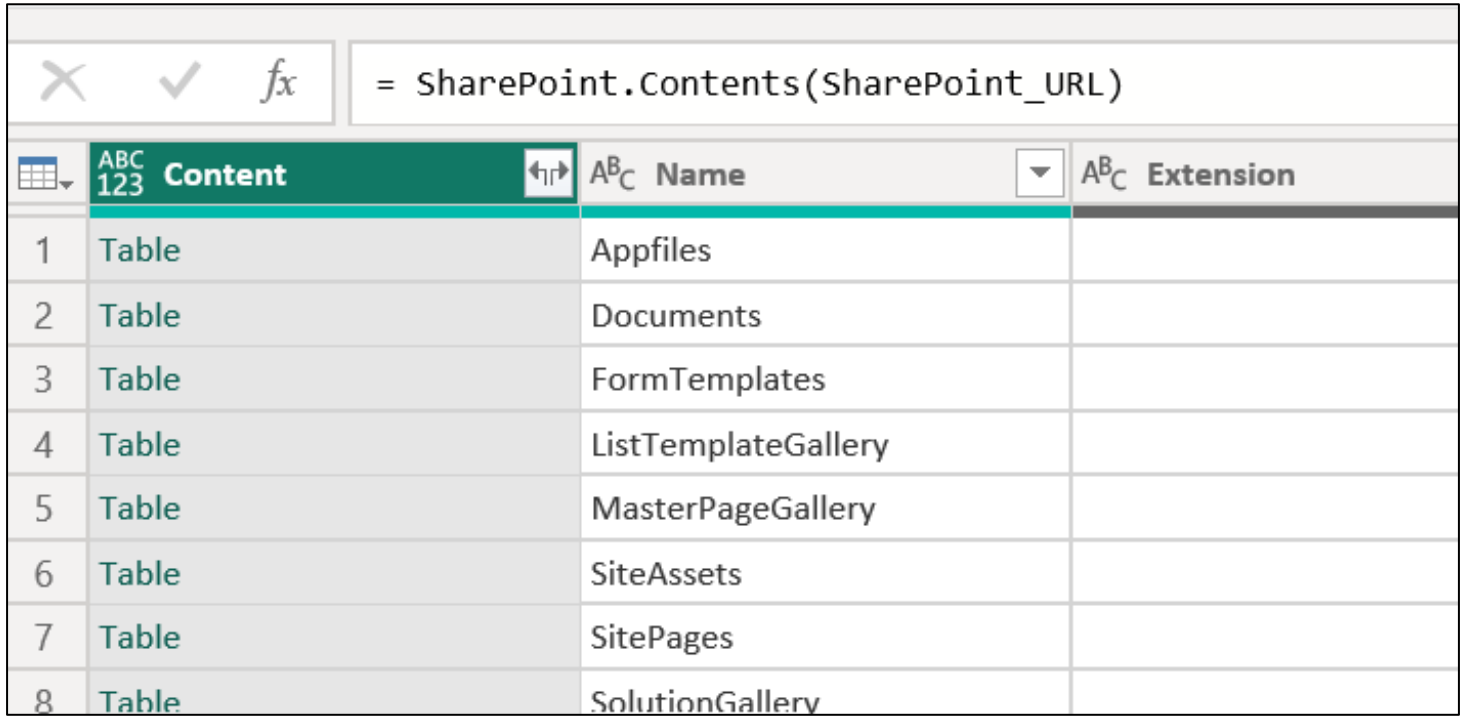
Pros

- Best suited for:
  - Documents of a single, consistent type.
  - Site has one Document Library
  - Combining files across Document Libraries

Cons

- Slower developer experience
  - Loads all file metadata
- Slower when > 100 files

SharePoint.Contents



SharePoint.Contents - Pros & Cons

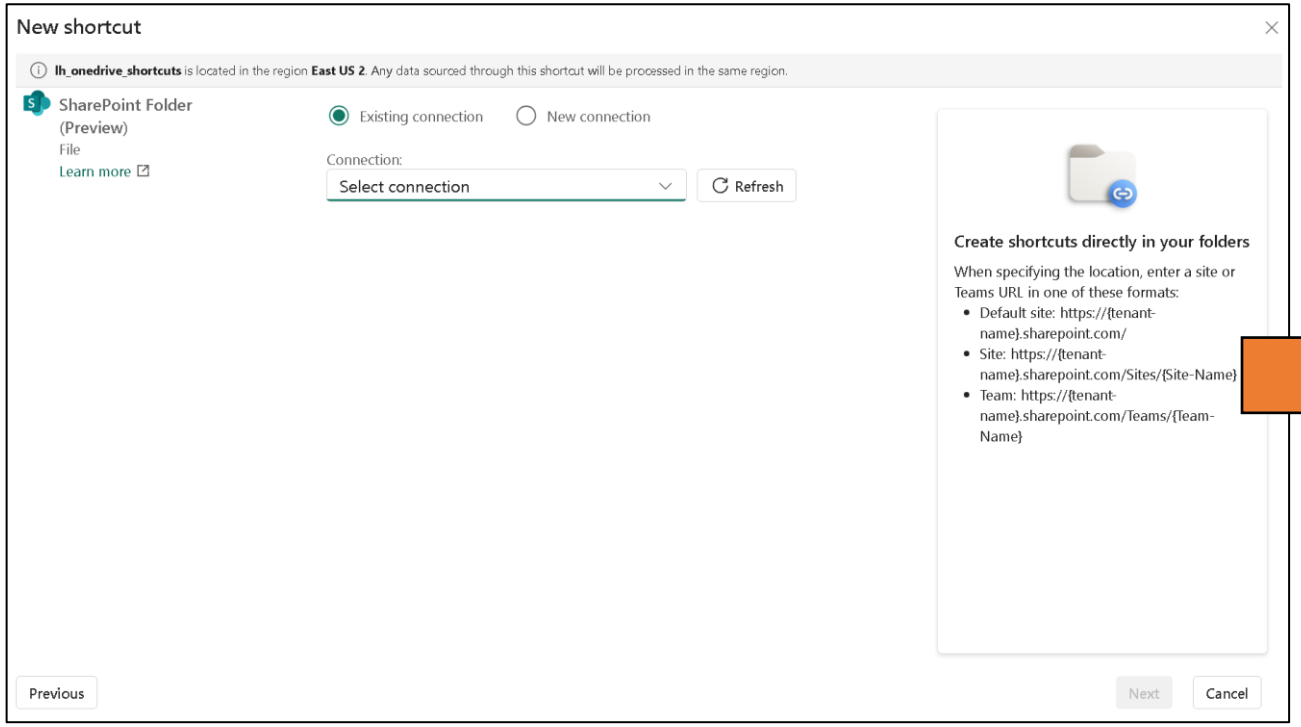
Pros

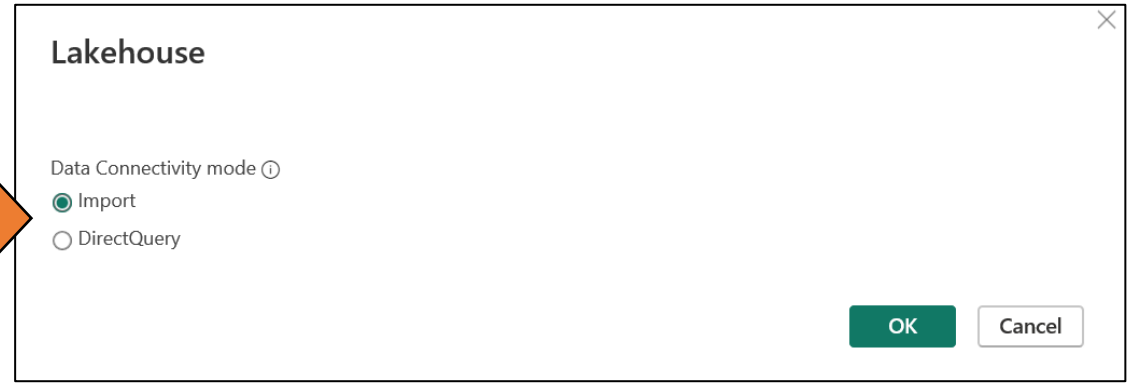
- Best suited for:
  - Sites with lots of files
  - Files in specific folders

Cons

- Need multiple tables if pulling multiple document libraries.

OneLake Shortcuts – SharePoint Folder





OneLake Shortcuts - Pros & Cons

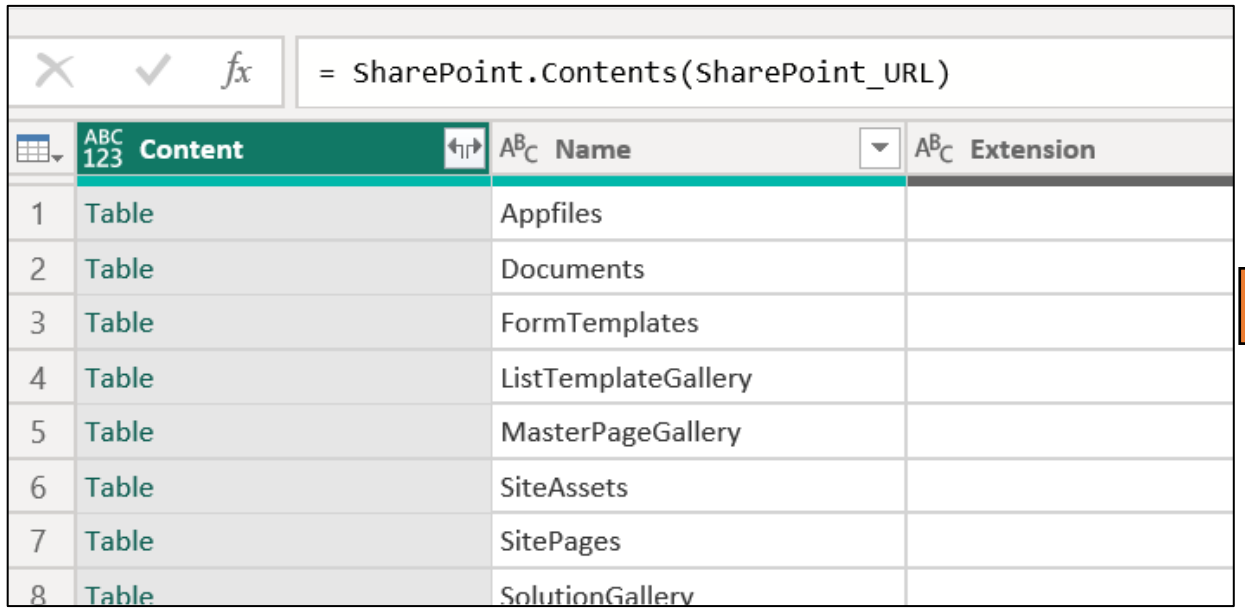
Pros

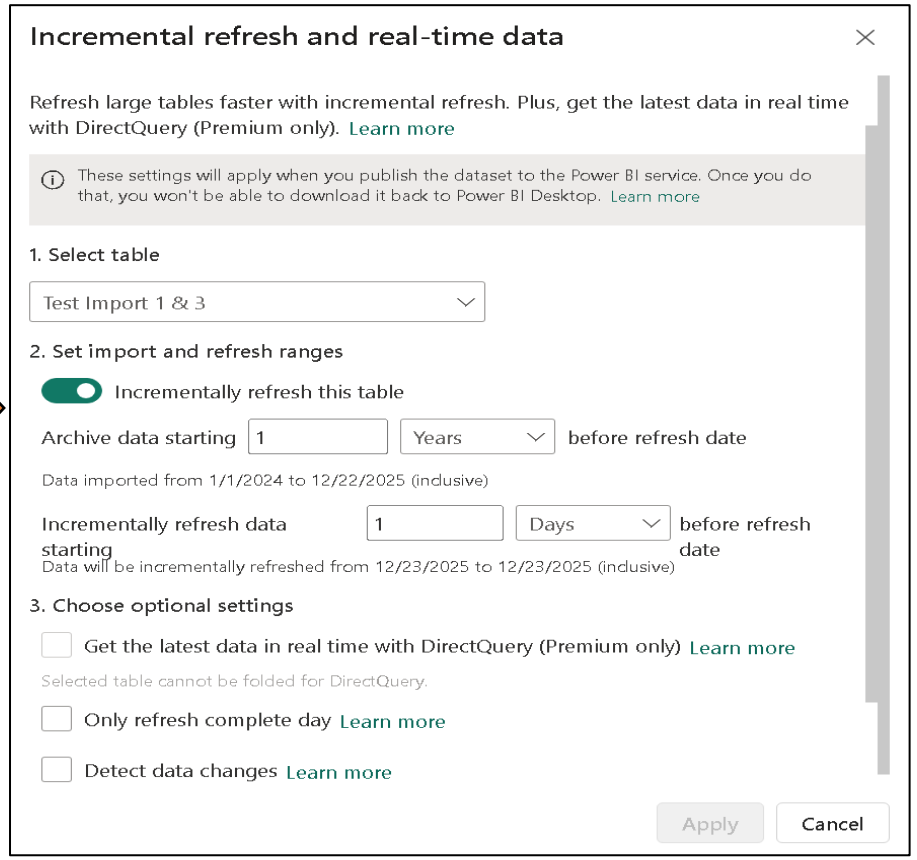
- No Manual Pipelines
- Updated every 2 minutes
- Allows you to process outside of the semantic model

Cons

- Need Fabric Licensing
  - Unsupported in GCC/DoD
- Slower if not processing outside of semantic model

SharePoint.Contents - Incremental





Results on a Fabric Capacity (> 1,000 Files)

Dataset	Median – Seconds to Refresh	Standard Deviation – Seconds to Refresh	Capacity Units	Number of Refreshes
One-Lake Shortcuts	568.2	154.9	5,139.57	10
SharePoint.Contents	461.9	173.0	2,613.52	10
SharePoint.Files	408.0	182.3	3,266.31	10
SharePoint.Contents - Incremental	68.6	155.4	204.57	10

SharePoint.Contents - Incremental - Pros & Cons

Pros

- Faster Reloads

Cons

- More Setup Steps

Document Libraries – Best Practices

- Use SharePoint.Contents for Large Libraries
- Implement Incremental
- Add Development Parameter
- Future
  - Consider Shortcuts

Thank you!

